Heat Treatment and Mechanical Characterization of Mud Bricks

A project report submitted

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Submitted by

Daksh Garg
2023mmb1410

Naveen Kumar
2023mmb1417

DEPARTMENT OF METALLURGICAL AND MATERIALS ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY ROPAR

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ABSTRACT

This project looks at developing and testing smaller mud bricks made from soil sourced locally. The main goal is to assess how different firing temperatures and holding times in a muffle furnace impact the bricks' strength, weight loss, and water absorption.By reducing the bricks to one-eighth of their original size by halving each linear dimension, we save materials and energy while gathering reliable data for real-world construction. We created 42 scaled-down mud bricks using a custom 25-gauge Galvanized Iron (GI) sheet metal mold. The locally sourced soil was screened to remove coarse particles and mixed with water to achieve the right texture. After being manually pressed and molded, the bricks were left to dry in the sun for about three days. This helped eliminate excess moisture and reduced the chances of cracking during firing. Controlled firing took place in a laboratory muffle furnace at two main temperatures: 900 °C and 1000 °C. We also tried firing at 800 °C initially, but this led to weak, under-fired bricks. For both main temperatures, we fired separate batches at five different holding times: 9, 12, 18, 24, and 36 hours. We recorded the weights before and after firing to measure weight loss. We then tested water absorption for the 36-hour fired batches according to IS 3495 (Part 2). We conducted compressive strength tests with a Compression Testing Machine (CTM) that has a capacity of 3000 kN at the Concrete Laboratory, IIT Ropar. Each scaled-down brick had a cross-sectional area of 6132 mm^2, and we applied the load at a rate of 14 MPa/min. We generated real-time graphs of load versus strength, and we noted the strength at which the first major crack occurred to indicate the initial load-bearing capacity of each brick. The results showed that firing temperature and holding time are crucial factors affecting the quality of fired mud bricks. The batch fired at 800 °C had an average compressive strength of only 2.96 MPa, which did not meet the minimum structural standards. Bricks fired at 900 °C reached a maximum average strength of about 8.07 MPa at 36 hours, classifying them as Second-Class bricks. The batch fired at 1000 °C performed best, with the 36-hour samples achieving an average compressive strength of 11.83 MPa, qualifying them as premium First-Class bricks suitable for heavy loads. Weight loss for all batches ranged from 6.55% to 7.13%, and water absorption values for successful batches were between 13.72% and 14.99%, all within the acceptable IS 3495 limit of 5% to 20%. This study confirms that firing at 1000 °C with a 36-hour soaking time provides the best results and supports the scaled-down experimental method as an effective and resource-saving approach for mud brick research.

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1. LITERATURE REVIEW

1.1 Introduction to Mud Bricks

Figure 1.1: Traditional mud brick construction kiln in India

Mud bricks are oldest construction material used in India especially in rural area. They are mainly made from soil, water and natural fibres such as straw or husk. Raw materials are easily available and manufacturing cost is low. Hence, mud bricks are widely used. They are also considered eco-friendly as they require less energy as compared to cement blocks and fired clay bricks. Another great advantage of mud bricks is their good thermal insulation property. It helps houses to be cool in summer and warm in winter. However, traditional mud bricks have some disadvantages despite these advantages. They are weaker than many modern building materials and are easily affected by water and environmental conditions.these problems have led many researchers to study the traditional mud brick manufacturing methods in order to understand their properties and improve their quality.

1.2 Soil Composition and Characteristics

The quality of mud bricks depends mainly on the composition of the soil used in their preparation. A good soil mix should generally have clay, sand and silt in equal proportions. Clay is a natural binder that helps hold the particles together, and sand reduces shrinkage and cracking when it dries. Silt improves the workability of the mix during molding. Many studies show a good mud brick soil to be around 15–30% clay and 50–70% sand with a small amount of silt and very little organic matter. If the clay content is too high the bricks may crack after drying because they shrink too much. Excess sand likewise reduces the bonding and makes the bricks weak. Before preparation, soil is often cleaned to remove stones and other unwanted particles. Then water is added to produce a homogeneous mixture with good plasticity. In traditional Indian methods, mixing is mainly done by hand or foot. In some areas straw or rice husk is also added to reduce cracking and improve bonding.

1.3 Traditional Mud Brick Manufacturing Process and Its Problems

The traditional mud brick manufacturing process in India is simple and mostly manual. The first step is to collect and clean suitable soil. Then the dirt is mixed with water to make workable mud paste. After mixing the wet mud is placed inside wooden or metal molds that gives the bricks a fixed shape and size. The molded bricks are taken out carefully and dried in the sun for few days. The dried bricks can then either be used in their natural state for building or fired in traditional kilns to make them stronger and more durable. This approach is low-cost but suffers from various practical issues. One of the biggest problems is the variation in soil composition from place to place which affects the strength and quality of the bricks. Manual mixing can also cause uneven moisture distribution, creating weak areas in the bricks. In sun drying, shrinkage cracks often appear on the surface because of the rapid evaporation of water. Mud bricks can absorb water easily and can be less durable in rainy conditions. The conventional methods of molding lead to the production of bricks of irregular dimensions sometimes and this creates problems in construction. Traditional kilns do not provide uniform heating and this results in bricks that are either over-burnt or under-burnt. These limitations have motivated researchers to keep researching traditional mud brick making methods to improve their performance and reliability.

1.4 Heat Treatment and Its Significance

Heat treatment is important for the improvement of properties of mud bricks. In traditional brick making, the bricks are often heated or fired in kilns after drying. As the bricks are heated, the moisture inside them is gradually removed and the soil particles become more compact and stable. The process increases the hardness, strength and durability of the bricks. Good heat treatment also reduces water absorption and improves resistance to environmental conditions. However, if the temperature is not well controlled, the overheating can cause thermal cracks and brittleness in the bricks. So, it is very important to study the effect of heat treatment on mud bricks to understand the effect of temperature on the performance and quality of mud bricks.

CHAPTER 2

OBJECTIVE OF THE WORK

2. Objective of the Work

- Design an efficient and resource-saving experimental setup by scaling down mud brick dimensions to half their original length, width, and height. This will create a test volume that is $1/8^{\text{th}}$ of a standard brick.
- Calculate dimensions and build a custom 25-gauge Galvanized Iron (GI) sheet metal mold to produce uniform, smaller mud bricks.
- Assess the physical strength of the scaled-down mud bricks during a 3-day sun-drying period before placing them in the furnace.
- Examine the effects of different firing temperatures by exposing batches of bricks to 800°C, 900°C, and 1000°C in a muffle furnace.
- Determine the impact of prolonged thermal soaking by keeping the specified firing temperatures for different times of 9, 12, 18, 24, and 36 hours.
- Analyze the final ceramic products by conducting tests for compressive strength, total weight loss during firing, and water absorption rates to find the best manufacturing parameters.

CHAPTER 3

EXPERIMENTAL PROCEDURE

3. Experimental Procedure

3.1 Mould Preparation

To accurately create the scaled-down bricks, three full-sized standard bricks were initially measured to establish an accurate average baseline.

$$L_{avg} = \frac{22.0 + 22.1 + 22.2}{3} = 22.1 \text{ cm}$$
$$B_{avg} = \frac{11.1 + 11.0 + 11.2}{3} = 11.1 \text{ cm}$$
$$H_{avg} = \frac{7.5 + 7.4 + 7.4}{3} = 7.43 \text{ cm}$$

These average dimensions were halved to calculate the target geometry for the $1/8^{\text{th}}$ volume experimental mould:

$$L = 11.05 \text{ cm}$$
$$B = 5.55 \text{ cm}$$
$$H = 3.72 \text{ cm}$$

The total required GI sheet dimensions were calculated as:

$$L_{\text{sheet}} = 11.05 + 3.72 + 3.72 = 18.49 \text{ cm}$$
$$B_{\text{sheet}} = 5.55 + 3.72 + 3.72 = 12.99 \text{ cm}$$

A final sheet size of **20** cm $\times$ **14** cm was selected to allow proper folding and overlap. A 25-gauge GI sheet was cut and folded into the mould using standard workshop techniques.

3.2 Mud Brick Preparation

The locally sourced soil was passed through a fine screen to remove stones and impurities. We gradually added water to achieve the right plasticity and proper mixing. Then, we manually pressed the prepared mud into the fabricated GI mold, creating a total of 42 scaled-down mud bricks for testing.

Figure 3.2: Prepared scaled down GI sheet mould

Figure 3.3: Manually Prepared 42 mud bricks

3.3 Sun Drying Phase

After molding, we arranged the bricks in an open area and let them dry naturally under sunlight for about 3 days. This process removed most of the free moisture and reduced the chance of cracking during furnace heating.

Figure 3.4: Sun drying of freshly moulded mud bricks

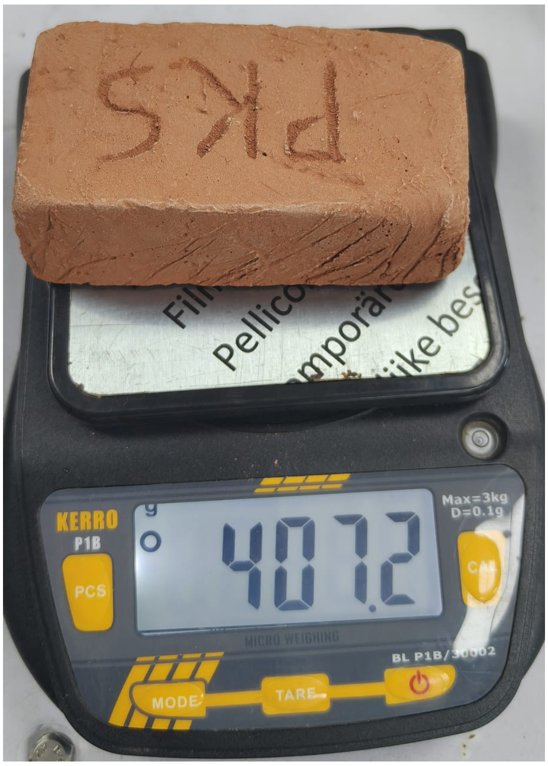
3.4 Controlled Firing Process

The controlled firing process took place inside a laboratory muffle furnace. First, all sun-dried bricks were weighed before firing to record their pre-firing weights. A preliminary firing trial happened at 800°C for 9 hours. However, the fired bricks were still weak and under-fired, showing that they did not vitrify enough. As a result, the remaining experiments were conducted only at 900°C and 1000°C. For both temperatures, separate batches were fired for 9, 12, 18, 24, and 36 hours. The 9, 12, 18, and 24-hour conditions used 3 bricks each, while the 36-hour condition used 6 bricks for both temperatures to allow for more testing. After firing, the furnace was allowed to cool slowly to room temperature while remaining closed to prevent thermal shock. Finally, the post-firing weights of all bricks were measured and recorded.

(a) 3 bricks were loaded inside the muffle furnace for controlled firing

(b) 6 bricks were loaded inside the muffle furnace for controlled firing

(c) weight of pre fired mud bricks



(d) weight of fired mud bricks

Figure 3.5: Controlled firing process stages inside the muffle furnace

3.5 Water Absorption Test

We conducted water absorption testing based on **IS3495 (Part 2)**, using the 36-hour fired samples at 900°C and 1000°C. We selected three bricks from each condition for the tests. First, we placed the brick samples in a hot air oven set to approximately 100°C to eliminate any residual moisture. Once fully dried, we cooled the specimens and recorded their dry weights. Next, we completely submerged the dried bricks in clean water for 24 hours. After immersion, we took the specimens out and carefully wiped them with a clean cloth to remove surface water. We immediately measured their saturated weights after wiping. We calculated the percentage of water absorption using:

$$\text{Water Absorption (\%)} = \frac{W_2 - W_1}{W_1} \times 100$$

where W_1 = dry weight before immersion and W_2 = wet weight after immersion.

(a) bricks were placed inside a hot air oven at 100°C to remove residual moisture

(b) Bricks were fully immersed in clean water for 24 hours

Figure 3.6: Water absorption test for fired mud bricks

3.6 Compressive Strength Testing

We performed compressive strength testing in the Concrete Laboratory of IIT Ropar with a **Compression Testing Machine (CTM)** that has a capacity of 3000 kN. To find the cross-sectional area of the scaled-down brick, we used the measured dimensions:

$$A = 110.5 \times 55.5 = 6132 \text{ mm}^2$$

The CTM machine applies loading at a rate of 14 MPa/min. Therefore, the required loading rate was:

$$\text{Load Rate} = \frac{6132 \times 14}{60} \approx 1430 \text{ N/s}$$

During testing, each specimen was positioned carefully between the compression platens for proper axial loading. The CTM software generated real-time load and strength graphs. Any sudden drops in the graph indicated the start of major cracks, and we recorded the strength at the first major crack initiation for analysis.

(a) Specimen positioned between plates of CTM machine

(b) data given to software of CTM

(c) CTM output graph (strength vs time)

Figure 3.7: Compressive strength testing of fired mud bricks

CHAPTER 4

RESULTS AND DISCUSSION

4. Result and Discussion

4.1 Color Observations After Firing

Significant visual and structural differences appeared in the mud brick samples after firing at different temperatures. The changes in color, surface texture, and crack formation provided important clues about oxidation and vitrification in the bricks during heat treatment.

4.1.1 Sample A (800°C)

The brick fired at 800°C showed a pale yellowish-brown sandy earth tone and had severe, deep surface cracks. At this lower temperature, there was not enough thermal energy for the iron compounds in the clay to oxidize properly and turn into stable iron oxide phases that give red bricks their color. The lack of sufficient heat also prevented effective vitrification of silica within the structure. Consequently, the bonding between particles remained weak, leading to extensive cracking and fragile behavior after firing.

4.1.2 Sample B (900°C)

The brick fired at 900°C exhibited a noticeable shift to a light terracotta or orange-red color. This temperature marked a critical transformation stage where the iron minerals in the clay began converting into hematite (Fe_2O_3), resulting in the characteristic red color of fired bricks. Some vitrification also started at this temperature due to limited silica melting, which improved the structural integrity of the specimens. While the brick appeared stronger and denser compared to the one fired at 800°C, minor surface cracks were still present.

4.1.3 Sample C (1000°C)

The brick fired at 1000°C developed a uniform deep terracotta red color over its entire surface. This intense red showed the successful oxidation of iron compounds into hematite (Fe_2O_3). At this temperature, significant vitrification occurred as silica phases softened and partially melted, creating strong internal bonds between the clay particles. As a result, the surface became smoother, denser, and largely free from major cracks. The better densification and reduced porosity suggested improved mechanical strength and durability of the fired brick.

Figure 4.8: Sample A : 800°C

Figure 4.9: Sample B : 900°C

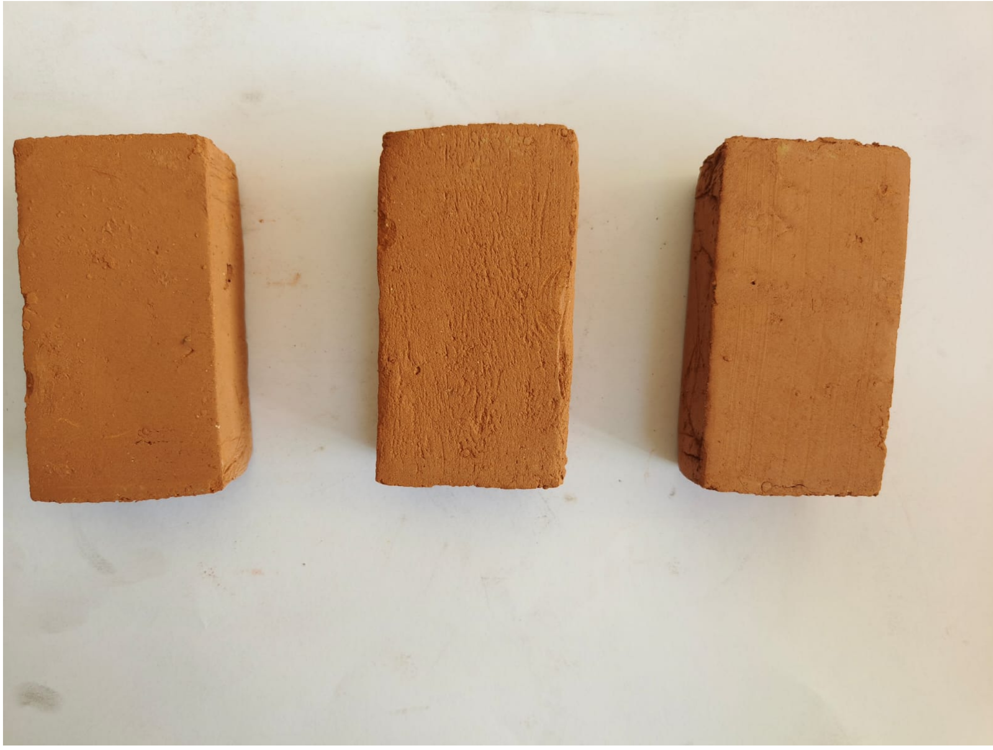


Figure 4.10: Sample C : 1000°C

4.2 Weight Loss Analysis

The % weight loss of the mud brick samples was calculated by comparing the weights before and after firing. The differences in weight loss related to firing temperature and duration provided valuable information about moisture removal, decomposition of volatile compounds, and densification of the clay during heat treatment.

Table 4.1: %Weight Loss Data for Mud Brick Specimens at Different Firing Conditions

S.No.	Temp. (°C)	Time (hours)	Avg. Dry Wt. (g)	Avg. Final Wt. (g)	Wt. Loss (g)	% Wt. Loss
1	900	9	427.30	399.30	28.00	6.55%
2	900	12	433.73	405.17	28.57	6.59%
3	900	18	433.07	404.30	28.77	6.64%
4	900	24	436.17	407.37	28.80	6.60%
5	900	36	418.55	390.35	28.20	6.74%
6	1000	9	435.73	404.67	31.07	7.13%
7	1000	12	412.67	383.67	29.00	7.03%
8	1000	18	414.50	385.97	28.53	6.88%
9	1000	24	420.70	392.57	28.13	6.69%
10	1000	36	431.33	402.17	29.17	6.76%

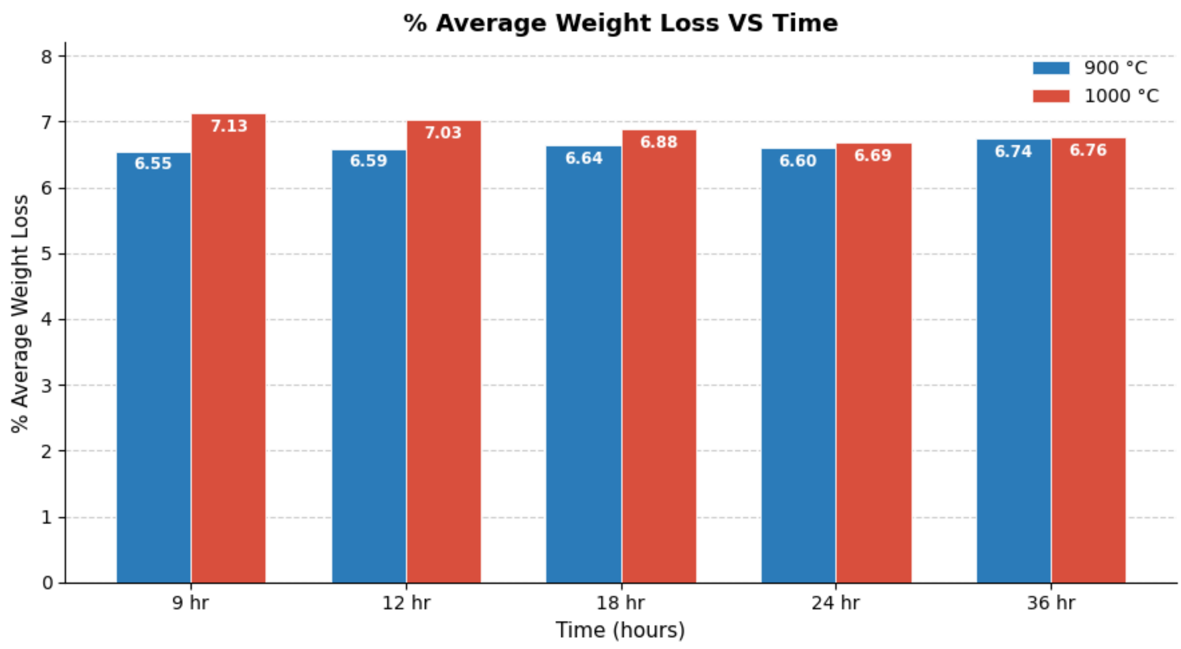


Figure 4.11: % Weight loss vs. holding time

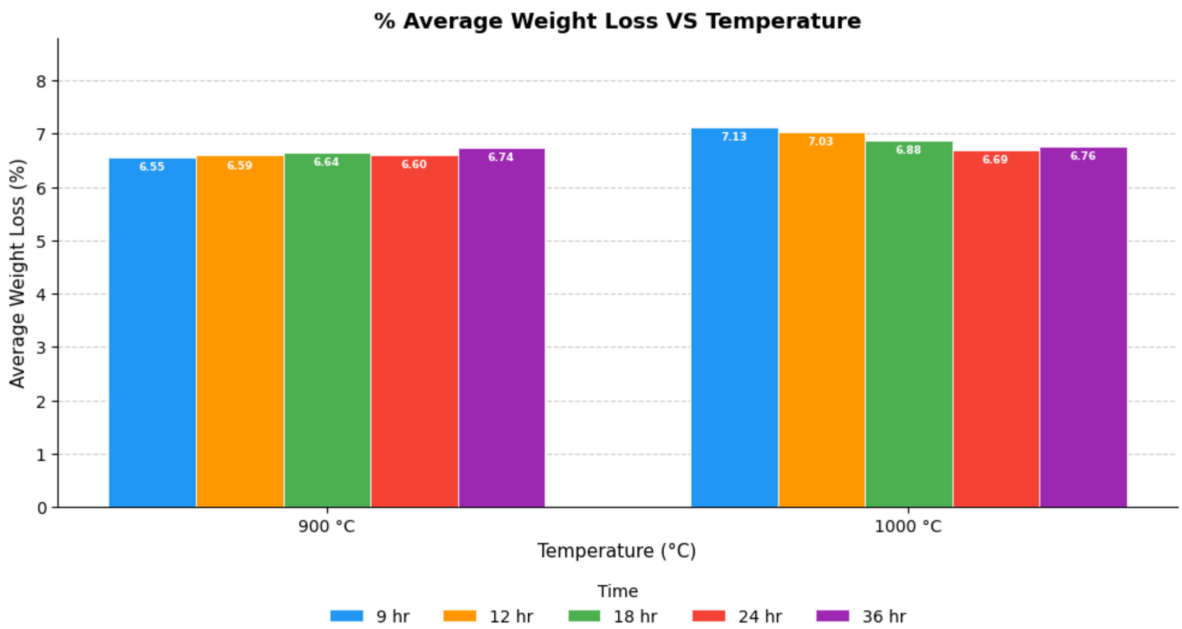


Figure 4.12: % Weight loss vs. Temperature

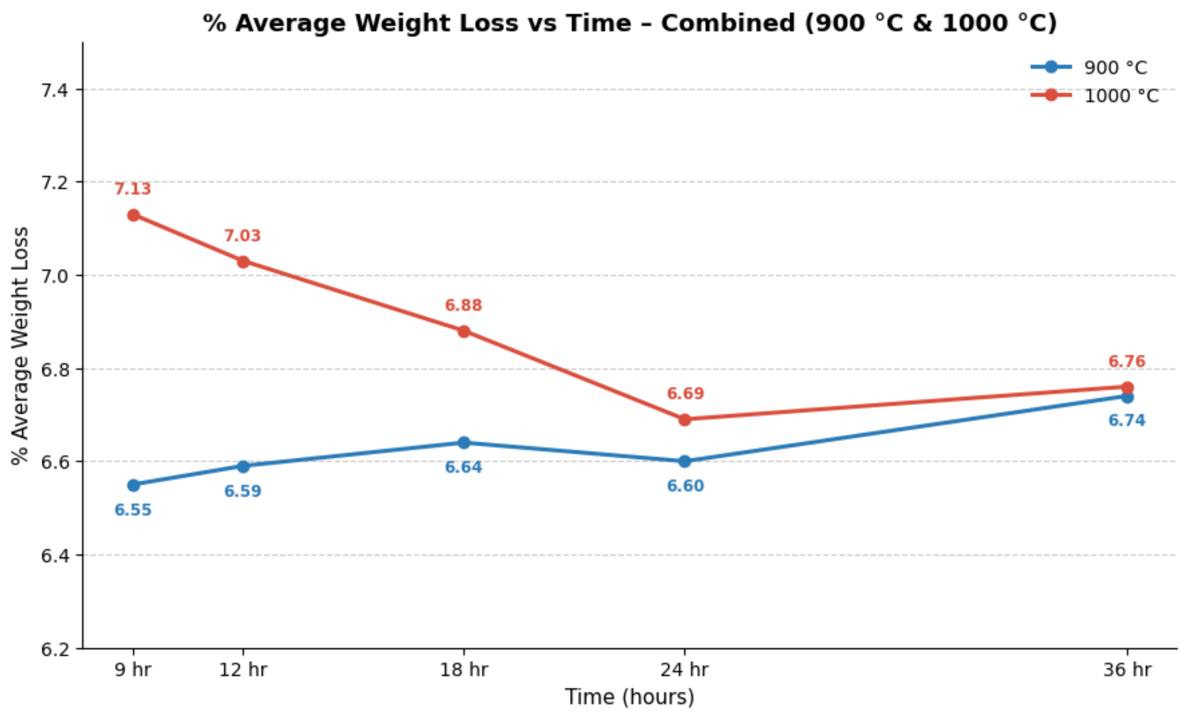


Figure 4.13: Comparative % weight loss at 900°C and 1000°C

4.2.1 Observations at 900°C

For the 900°C batch, the average weight loss remained fairly stable throughout the firing durations, showing a gradual increase from about 6.55% at 9 hours to nearly 6.74% at 36 hours. This steady trend indicates that thermal decomposition reactions happened gradually during heating. The slow increase in weight loss can be attributed to the gradual removal of chemically bound water, breakdown of carbonate phases, and burning off small amounts of organic matter in the clay. The absence of sudden changes suggests that firing at 900°C encouraged relatively stable thermal reactions and controlled densification of the brick.

4.2.2 Observations at 1000°C

The samples fired at 1000°C initially had a much higher weight loss of about 7.13% at the 9-hour mark. This large reduction in weight confirmed that the higher firing temperature sped up the removal of moisture, volatile compounds, and chemically bound elements from the clay. However, rather than following the expected trend of continuously increasing weight loss, the later batches showed a slight drop, falling to about 6.69% at 24 hours and stabilizing around 6.76% at 36 hours.

4.2.3 Discussion of 1000°C Data Variation

This slight deviation from the expected trend is mainly due to experimental and environmental factors. There was a noticeable gap between the firing times of the initial batches (9 and 12 hours) and the later ones (18, 24, and 36 hours), during which the samples were exposed to the open laboratory environment. This extra atmospheric exposure allowed for additional natural air drying, which lowered residual moisture content before firing, making the percentage weight loss appear lower. Inconsistencies in manual fabrication concerning compaction density, moisture distribution, and water-to-clay ratio also added to the observed variations.

4.2.4 Comparative Discussion

Despite these intermediate variations, the final percentage weight loss values for both the 900°C and 1000°C batches were very close after extended soaking durations. At 24 and 36 hours, both conditions remained within a narrow range of about 6.69% to 6.76%. This suggests that after prolonged soaking, the clay matrix reached a state of thermal equilibrium and maximum densification, regardless of initial moisture differences.

4.3 Water Absorption Analysis

The water absorption behavior of the fired mud brick specimens was analyzed according to the results obtained from immersion testing performed as per **IS 3495 (Part 2)**.

4.3.1 Data Observation

Based on the experimental results, the percentage water absorption values for all specimens fired for 36 hours at 900°C and 1000°C were found to lie within the range of approximately 13.72% to 14.99%.

Table 4.2: % Water Absorption Test Data for Mud Bricks

Sr. No.	Brick No.	Temp. (°C)	Holding Time (hours)	Fired Wt. (g)	Wt. After Absorption (g)	% Water Absorption
1	M3	900	36	359.50	409.60	13.94%
2	M4	900	36	377.50	434.10	14.99%
3	M6	900	36	403.30	462.70	14.73%
4	N3	1000	36	411.90	469.70	14.03%
5	N5	1000	36	399.50	454.30	13.72%
6	N6	1000	36	396.70	452.50	14.07%

4.3.2 Effect of Temperature

The specimens fired at 1000°C had a slightly lower average water absorption rate of nearly 13.9% compared to the 900°C specimens, which had an average absorption around 14.5%. This decrease is linked to improved vitrification at the higher temperature; partial melting of silica phases helps seal pores and densify the material, limiting water penetration and retention.

4.3.3 Validation with Indian Standard Limits

According to IS 3495, a high-quality first-class brick should have a water absorption rate between 5% and 20% of its dry weight after 24 hours of immersion. All experimental specimens met this requirement, with measured values comfortably within about 13% to 15%, confirming they meet quality standards for structural use.

4.3.4 Porosity and Mortar Bonding Characteristics

The absorption range of about 13% to 15% shows a good balance between densification and controlled surface porosity. The specimens were dense enough to handle compressive loads while still retaining enough surface porosity to absorb wet cement mortar, improving mechanical interlocking and bonding strength in masonry applications.

4.3.5 Consequences of Excessively High Water Absorption

If water absorption had exceeded 20%, it would indicate highly porous, poorly fired, and structurally weak bricks. These bricks would soak up excessive moisture from the environment, leading to physical degradation, internal dampness, reduced mechanical durability, and vulnerability to efflorescence.

4.3.6 Consequences of Extremely Low Water Absorption

On the other hand, absorption below 5% would suggest severe over-firing and excessive vitrification. Although very vitrified bricks may be hard, their smooth, non-absorbent surfaces hinder bonding with cement mortar, making them unsuitable for ordinary structural wall construction.

4.4 Compressive Strength Analysis

4.4.1 Data Observations

Table 4.3: Average Compressive Strength Data for Mud Bricks at Different Firing Conditions

Sr. No.	Temp. (°C)	Holding Time (hours)	B1 (MPa)	B2 (MPa)	B3 (MPa)	Avg. Strength (MPa)
1	800	9	3.12	2.97	2.80	2.96
2	900	9	5.42	3.52	4.73	4.56
3	900	12	6.47	6.34	5.99	6.27
4	900	18	7.52	6.79	8.15	7.49
5	900	24	8.90	8.23	5.97	7.70
6	900	36	8.19	7.81	8.21	8.07
7	1000	9	7.21	5.59	8.02	6.94
8	1000	12	8.17	8.59	6.81	7.86
9	1000	18	7.88	6.98	11.89	8.92
10	1000	24	9.96	10.64	9.71	10.10
11	1000	36	11.81	11.16	12.51	11.83

- **Effect of Temperature:**Firing temperature is the main factor affecting structural strength. The 800°C batch proved severely under-fired, yielding only 2.96 MPa. The 1000°C batch showed much higher strength at all time intervals compared to the 900°C batch.

- **Effect of Holding Time:** Longer soaking time steadily increases structural integrity. In the 1000°C batch, compressive strength rose from 6.94 MPa at 9 hours to a peak of 11.83 MPa at 36 hours, confirming that thermal energy needs ample time to fully penetrate the core and cure the material.

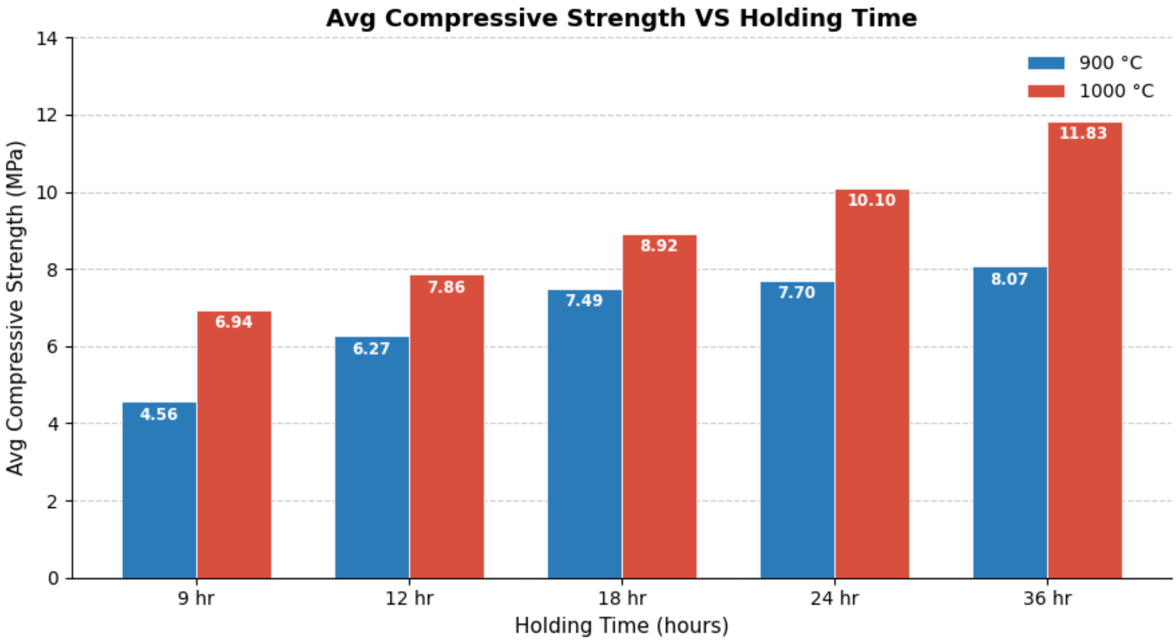


Figure 4.14: Compressive strength vs. holding time

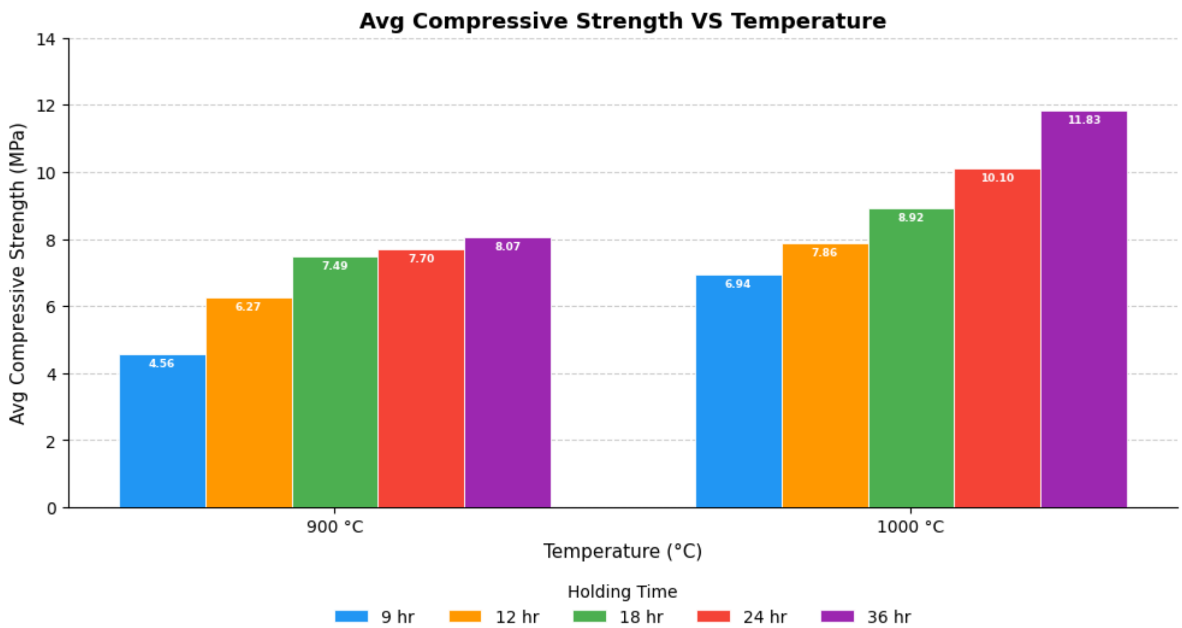


Figure 4.15: Compressive strength vs. Temperature

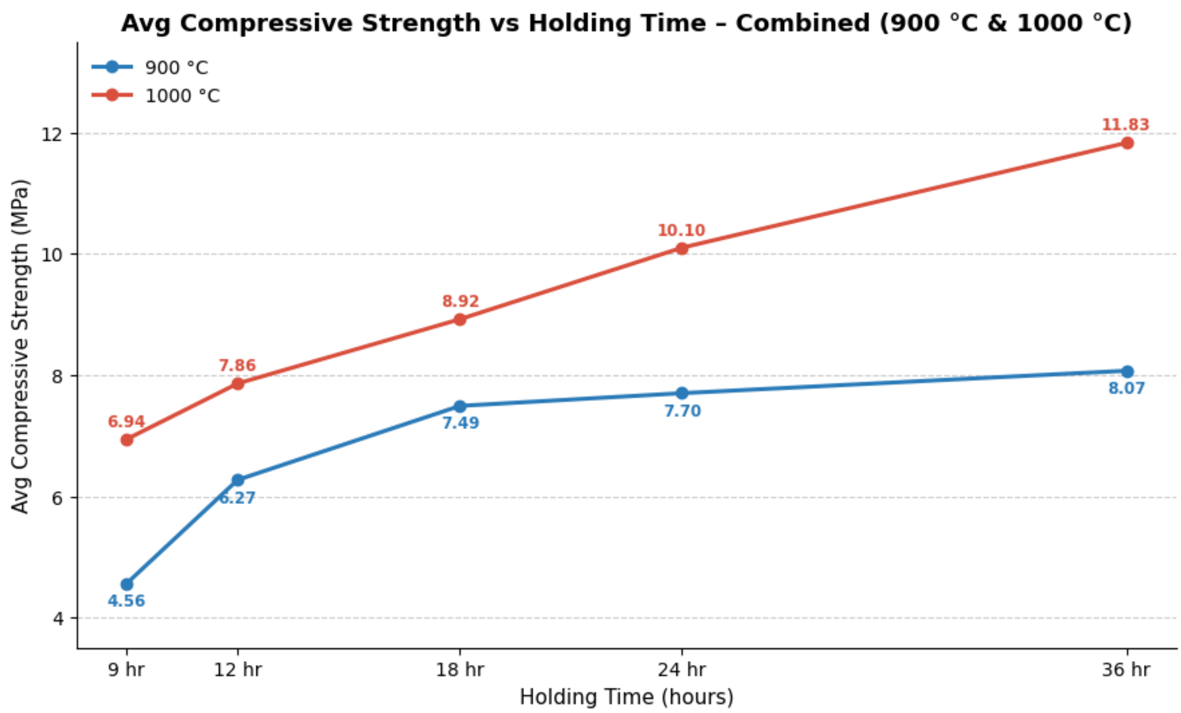


Figure 4.16: Comparative compressive strength at 900°C and 1000°C

4.4.2 Scientific Reasoning for Strength Increase

- **Vitrification (paritial Silica Melting):** The surge in strength at 1000°C results from sufficient thermal energy causing the silica and natural fluxing agents within the clay to partially melt, forming a viscous liquid glass phase inside the brick.
- **Densification and Pore Sealing:** As the brick cools, this melted silica hardens into a strong internal binder, filling and sealing tiny air voids between clay particles. This significant reduction in porosity results in a dense, solid matrix that can withstand heavy compressive loads.

4.4.3 Validation Against Indian Standards (IS 1077)

- **800°C Batch ($<$ 3.5 MPa):**An average strength of just 2.96 MPa falls below the minimum threshold, failing to qualify even as Third-Class bricks and being structurally unfit for construction.
- **900°C Batch (Peak $\sim$8.07 MPa):**Samples fired for 18, 24, and 36 hours exceed the 7.0 MPa minimum, categorizing them as Second-Class bricks suitable for internal or non-load-bearing partition walls.
- **1000°C Batch (Peak $\sim$11.83 MPa):**The 24-hour (10.10 MPa) and 36-hour (11.83 MPa) batches comfortably exceed the 10.5 MPa mark, classifying them as premium First-Class bricks suitable for heavy load-bearing exterior structures.

4.5 Defects

4.5.1 Breakage Due to Large Particles

- **Stress from Uneven Shrinkage:**Large, uncrushed particles, like stones or pebbles, do not shrink during drying. As the surrounding wet clay shrinks around these solid inclusions, it creates severe internal tension that can lead to hidden cracks.
- **Thermal Expansion Mismatch:** During high-temperature firing, stone and clay expand and cool at different rates. This uneven thermal movement acts like an internal wedge, causing the brick to crack from within.
- **Catastrophic Structural Failure:**These internal cracks create major weak points, leading to a loss of compressive strength and an increased chance of snapping or crumbling at the site of the large particle under pressure.

Figure 4.17: Brick fracture due to large particle inclusion

CHAPTER 5

CONCLUSIONS / SUMMARY

5. Conclusions / Summary

The main goal of this experiment was to investigate how firing temperature and time affect compressive strength, weight loss, and water absorption of scaled-down mud bricks. Based on the experimental results and observations, the following conclusions can be made:

- **Optimal Firing Temperature:**A firing temperature of 1000°C was optimal. At this temperature, the internal silica phases melted and vitrified, creating a dense and strong brick. Samples fired at 1000°C for 24 and 36 hours achieved compressive strengths of 10.10 MPa and 11.83 MPa respectively, qualifying them as premium First-Class bricks suitable for heavy load-bearing structures.

- **Minimum Viable Temperature:** Firing at 900°C resulted in moderate outcomes. While it allowed for proper iron oxidation (creating a red color), vitrification was not complete. These bricks reached a peak strength of about 8.07 MPa, classifying them as Second-Class bricks. In contrast, 800°C was insufficient, resulting in extremely weak bricks (2.96 MPa) that did not meet structural standards.

- **Crucial Role of Holding Time:** Extended soaking time in the furnace is as crucial as peak temperature. Across all test batches, increasing the holding time steadily enhanced structural integrity. Longer heating lets the thermal energy penetrate the brick's core completely, ensuring uniform internal curing and maximum pore sealing.

- **Water Absorption Standards Met:**The water absorption rates for the successful batches at 900°C and 1000°C ranged from 13.72% to 14.99%. This fits well within the 5% to 20% range set by Indian Standard IS 3495. It ensures the bricks are porous enough to bond effectively with cement mortar, while being dense enough to withstand significant moisture damage.

- **Importance of Soil Screening:** During mud preparation, large, uncrushed particles, like stones, must be avoided. These impurities cause serious thermal and physical mismatches during firing, which can lead to major structural failure under pressure.

- **Validity of the Scaled-Down Approach:** Using an experimental setup that was scaled down to $1/8^{\text{th}}$ the volume of a standard brick proved very successful. It saved raw materials and furnace energy while still producing clear, uniform, and reliable data. This data can be used to predict how full-sized bricks will behave in real-world construction.

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